Distributed Edge Anomaly Detection: Architecture & Performance Evaluation

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1. Executive Summary

This evaluation report presents a comprehensive empirical assessment of the Distribut
The primary purpose of this study is to determine whether decentralized, local infere
The findings confirm that localized inference with periodic parameter synchronization

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2. Background and Problem Statement

Modern industrial automation depends heavily on reliable, real-time status reporting
However, centralized collection models suffer from three structural failures:

1. **Network Saturation and High Egress Costs:** Transmitting raw 100Hz vibration and
2. **Delayed Fault Isolation:** When a link experiences transient degradation, alert
3. **Privacy and Compliance Boundaries:** Direct transmission of raw industrial senso

Consequently, there is an urgent engineering requirement for edge-native anomaly dete

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3. Engineering Objectives

The evaluation was structured around four measurable technical objectives:

- **Objective 1 (Detection Accuracy):** Achieve an anomaly detection Area Under the R
- **Objective 2 (Throughput Optimization):** Reduce raw outbound data transmission vo
- **Objective 3 (Latency Guarantees):** Maintain 99th-percentile (p99) anomaly detect
- **Objective 4 (Partition Tolerance):** Maintain autonomous operational continuity f

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4. Evaluation Methodology

4.1 Testbed Architecture

The evaluation testbed comprised three distinct clusters of edge hardware representin
- **Cluster Alpha (Tier-1 Gateways):** 40x Raspberry Pi Compute Module 4 (Quad-core A
- **Cluster Beta (Tier-2 Micro-Gateways):** 50x BeagleBone Black Industrial (Single-c
- **Cluster Gamma (Industrial PLCs):** 30x Siemens SIMATIC IOT2050 (Dual-core ARM Cor
4.2 Workload Simulation & Fault Injection
A deterministic synthetic telemetry generator was developed to stream multidimensiona
- **Anomaly Type A (Sudden Spike):** 5x to 10x deviation lasting 100ms to 500ms (bear
- **Anomaly Type B (Drift):** Linear degradation of temperature/vibration over 4 hour
- **Anomaly Type C (Frequency Shift):** Harmonic distortion in 50Hz/60Hz frequency sp
- **Anomaly Type D (Intermittent Drop):** Packet loss and signal attenuation across s
4.3 Baseline Comparisons
The proposed DEAD architecture was evaluated against two industry standards:
1. **Baseline 1 (Centralized Cloud Ingestion):** Unfiltered compression and immediate
2. **Baseline 2 (Local Thresholding):** Static upper and lower control limits (3-sigm
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5. Experimental Data and Observations
Telemetry metrics were captured continuously across all 120 nodes. Table 1 summarizes
Table 1: Ingestion & Telemetry Summary Statistics
Metric Dimension	Cluster Alpha	Cluster Beta	Cluster Gamma	Combined Total
Active Node Count	40	50	30	120
Total Records Captured	5,840,000	4,920,000	3,440,000	14,200,000
Injected Anomaly Events	620	710	440	1,770
True Positives (TP)	598	642	412	1,652
False Positives (FP)	24	48	19	91
False Negatives (FN)	22	68	28	118
Mean CPU Utilization (%)	14.8%	38.6%	22.4%	26.1%
Memory Footprint (MB)	48.2 MB	36.4 MB	44.1 MB	41.8 MB

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6. Technical Analysis
6.1 Accuracy and Precision Tradeoffs
The experimental results demonstrate that local inferencing models achieved superior
6.2 Bandwidth and Data Reduction
Under the baseline cloud streaming model, each node generated an average of 42.6 MB o
6.3 Resource Utilization and Flash Wear
A critical risk factor observed on Cluster Beta (BeagleBone Black with 512MB RAM) was
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7. Comparative Results
Table 2 compares the core operational benchmarks across the three evaluated systems.
Table 2: Benchmark Comparison Matrix
| Architectural Approach | F1-Score | Detection Latency (p99) | Egress Bandwidth / Da
|---|---|---|---|---|
| Baseline 1: Cloud Streaming | 0.894 | 840ms | 1,022.4 MB | 0 Hours (Fails on partit
| Baseline 2: Static Thresholds | 0.732 | 12ms | 144.0 MB | 72+ Hours |
| Proposed: DEAD Architecture | 0.942 | 42ms | 268.2 MB | 72+ Hours |
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8. Strategic Recommendations
Based on empirical data gathered across the 120-node deployment, the engineering team
1. **Deploy Model Quantization for Tier-2 Nodes:** Convert float32 model weights to i
2. **Mandate In-Memory Ring Buffers for Offline Storage:** Require all edge logging p
3. **Implement Dual-Timescale Drift Compensation:** To eliminate Type B (slow drift)
4. **Automate Over-The-Air (OTA) Model Retraining:** Schedule weekly model parameter
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9. Conclusion
The evaluation demonstrates that the Distributed Edge Anomaly Detection (DEAD) framew